Synthetix’26 Electrical Stream Problem Statement

Modern autonomous and industrial systems frequently operate in electrically hostile environments containing switching power electronics, motor drives, RF emitters, and broadband electromagnetic interference. In such conditions, sensor outputs are often contaminated by multiple deterministic interference sources and noise, making reliable signal extraction a non-trivial analog design challenge.

In this challenge, teams must design an **analog-only signal conditioning module** that extracts a specified usable signal component from a standardized composite analog input while maintaining stability, repeatability, and robustness under interference.

This is a **pure analog design challenge**. Digital signal processing, microcontrollers, and programmable devices are strictly prohibited.

Scenario:

An autonomous inspection platform receives data from a wired sensor module operating inside an industrial environment.
The sensor produces a **composite analog signal** consisting of useful information embedded within multiple interference components and broadband noise.

The sensor module itself is treated as a **black box**. Teams are not allowed to modify or inspect its internal implementation.

Your task is to **separate and condition the usable signal component** such that it clearly dominates the output signal.

Objective:

Design, simulate, and validate an **analog signal conditioning circuit** that:

- Accepts a standardized composite analog input from a black-box source
- Suppresses interference and noise
- Produces a stable output dominated by the usable signal component
- Allows smooth, continuous manual adaptation during operation

Standardised input: (blackbox)

All teams will be provided with a **LTSPICE-compatible black-box subcircuit** that generates the composite input signal.

Signal Components (Internally Generated)

The composite signal includes the following primary deterministic components:

Component A – Low-Frequency Drift

- Waveform: Sine
- Frequency: 600 Hz ±2%
- Amplitude: 0.8 Vpp

Component B – Usable Information Signal

- Waveform: Sine
- Frequency: **6 kHz ±1%** *(primary evaluation mode)*
- Amplitude: 0.4 Vpp

This component must dominate the output after conditioning.

Component C – Structured Interference

- Waveform: Square wave
- Frequency: 18 kHz ±3%
- Amplitude: 1.2 Vpp

Component D – High-Frequency Interference

- Waveform: Sine
- Frequency: 55 kHz ±5%
- Amplitude: 2.0 Vpp

Component E – Broadband Noise

- Type: White noise
- RMS amplitude: 150 mVrms
- Bandwidth: ≥ 100 kHz

Along with linear components,

The source exhibits internal nonlinear dynamics that may alter the spectral characteristics of the usable component under varying signal conditions.

Composite Constraints

- Maximum instantaneous amplitude: ±2.5 V
- DC offset: ≤ ±50 mV
- Source impedance: 50 Ω

The internal implementation of the black-box source is **not accessible** and must not be modified.

Functional Requirements:

i) Output Signal Dominance:

- The output must be **clearly dominated by the usable signal component (6 kHz)**.
- All other spectral components must be significantly attenuated.
- Performance must be demonstrated using **frequency-domain (FFT) analysis**.

ii) Manual Adaptation interface:

The circuit must include at least one **physical manual control element** (e.g., potentiometer) that affects signal conditioning behavior.

Requirements:

- Continuous, smooth adjustment
- No abrupt switching or dead zones
- Predictable and repeatable response
- Usable while the circuit is operating

iii) Stability and Transient Behaviour:

- No sustained oscillations or runaway behavior
- No clipping or saturation under normal operation
- Output must remain continuous during control adjustment
- Transient excursions must settle smoothly without instability

iv) Noise and interference suppression Verification:

Teams must provide:

- Input vs output FFT plots
- Quantitative evidence of interference and noise reduction relative to the usable signal

Simulation tools such as LTspice, QSpice, MATLAB, Python, or GNU Octave may be used for analysis.

v) Calibration and repeatability:

- Teams must describe a repeatable method to correlate control position with output behavior.
- The relationship between control adjustment and output response must be consistent across simulations.

Constraints:

Not allowed –

- Microcontrollers
- DSP techniques
- Logic gates
- Programmable or adaptive ICs
- Dedicated filters or AGC ICs

Allowed-

- Op-amps
- Resistors, capacitors, Inductors
- Diodes
- BJTs/MOSFETs

Simulation Requirements:

- Circuit must be fully simulated using SPICE-based tools
- Both transient and frequency-domain behavior must be demonstrated

- The provided black-box source must be used for validation

BONUS challenges (Optional):

i) Hidden Operating Mode Robustness

During evaluation, judges may switch the usable signal frequency to **4 kHz or 8 kHz** using the black-box source.

- This will **not penalize** base scoring
- Used only for bonus points or tie-breaking
- Teams may detect frequency via analysis and adapt manually

ii) Signal Quality Indicator

Provide a secondary analog output indicating signal purity or dominance (e.g., envelope-derived voltage).

iii) PCB design:

Translate the validated analog signal conditioning design into a **compact, low-noise PCB layout** while preserving signal integrity, stability, and predictable behavior.

- Maximum PCB area: **≤ 50 cm²**
- Layer count: **2-layer maximum**
- Through-hole or SMD allowed
- No shield cans or external filtering modules

Deliverables

- PCB layout files (KiCad / Altium / EasyEDA, etc.)
- Board outline and layer views
- Annotated screenshots highlighting key layout decisions
- Short explanation (≤ 1 page) of grounding and noise strategy